Evaluation Protocol, Checkpointing, and Calibration Shape Apparent Multi-Task Gains in ADMET Prediction

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Abstract

We examine whether apparent multi-task gains in ADMET prediction are robust to evaluation protocol, probability calibration, and controls for cross-task leakage and training design. In a leakage-controlled diagnostic on eight binary ADMET endpoints from the Therapeutics Data Commons, we compare multi-task (MTL) with architecture-matched single-task (STL) models under global three-way molecule allocation (70%/10%/20% train/calibration/test) under grouped-random and scaffold-grouped protocols, with five split instances and three seeds. Under the prospectively frozen fixed-final checkpoint rule, MTL assigns lower raw log loss on all eight endpoints under both protocols, with a larger contrast under scaffold-grouped evaluation (protocol contrast $\boldsymbol{\Gamma = +0.081}$, split-instance bootstrap 95% CI $[\mathbf{+0.048, +0.112}]$). A validation-selected checkpoint sensitivity attenuates the absolute MTL–STL contrast under both protocols but preserves a positive protocol interaction (split-level $\boldsymbol{\Gamma}$ positive in all five split instances under every checkpoint rule). Under the frozen fixed-final rule, strict per-model temperature calibration—each temperature fitted on the same trained model's own calibration partition and applied to its own test partition—attenuates the raw-probability advantage and the protocol interaction to near-zero point estimates, consistent with the observed gains primarily reflecting temperature-correctable differences in raw logit scale. Compute-matched, label-permutation, and pooled-pretraining controls indicate that the raw gain is not reproduced by additional single-task updates and does not require genuine

auxiliary-label structure, and the protocol interaction survives full label permutation. A matched-scale graph neural network sensitivity analysis reproduced the positive direction of the protocol interaction (split-instance bootstrap interval including zero), and no systematic novelty dependence was found under either protocol. Reported multi-task advantages are therefore conditional on evaluation protocol and should be accompanied by calibration and training-rule diagnostics.
Contribution: We introduce a leakage-controlled framework for quantifying protocol-dependent MTL–STL proper-score contrasts in ADMET prediction and show that the apparent gain is larger under scaffold-grouped evaluation but that, under the frozen fixed-final rule, strict per-model temperature calibration attenuates it to near-zero point estimates. Compute-matched and label-permutation controls further show that the raw gain is not reproduced by additional single-task updates and should not be interpreted as evidence of semantic cross-task label transfer.

Keywords: multi-task learning, ADMET prediction, evaluation protocol, calibration, benchmarking

1 Introduction

Multi-task learning (MTL) is widely used in ADMET (absorption, distribution, metabolism, excretion, and toxicity) modeling, and several studies report that multi-task models improve over matched single-task (STL) baselines Jiang et al. (2025); Shahid et al. (2025). At the same time, the molecular machine-learning community has accumulated evidence that evaluation protocols materially shape conclusions: out-of-distribution (OOD) split protocols such as scaffold or cluster splits change absolute performance and its rankings Fooladi et al. (2025); Guo et al. (2025), and recent benchmarks in neighboring fields have shown that strong model claims can depend on evaluation details Ahlmann-Eltze et al. (2025); Wei et al. (2026).

Prior studies largely report MTL–STL comparisons under a single split protocol. It remains unclear whether the paired advantage itself changes across protocols, whether proper-score gains survive model-specific recalibration, and whether those gains require genuine auxiliary-task label information rather than shared optimization and auxiliary-input exposure. To our knowledge, prior work has not jointly characterized the paired MTL–STL contrast under leakage-controlled global molecule partitions, strict same-model recalibration, and mechanistic controls of the training design.

In this paper we report a leakage-controlled diagnostic whose final confirmatory plan was frozen before the corrected three-way rerun of task-balanced hard-sharing MTL on eight binary ADMET endpoints from the Therapeutics Data Commons Huang et al. (2021). The contributions are: (i) a paired MTL–STL protocol-contrast estimand evaluated under global molecule allocation with no cross-task identity exposure, under two valid protocols (grouped random and scaffold-grouped); (ii) a strict same-model calibration decomposition showing that the raw proper-score gain is attenuated to near-zero by per-model temperature rescaling; (iii) compute-matched (STL-8×), label-permuted, and pooled-pretraining controls that narrow the plausible explanations to effects associated with concurrent shared-trunk optimization and auxiliary-input exposure rather than semantic label transfer or additional target-task updates; and (iv) a matched-scale GNN sensitivity analysis together with a split-instance adequacy analysis. Endpoint-scale manipulation (downsampling) and the continuous novelty analysis are reported as secondary analyses.

The paper is organized as follows. Section 2 positions the work against prior studies of ADMET MTL, split protocols, and benchmark reliability. Section 3 defines the estimand hierarchy and the leakage-controlled design. Section 4 reports the protocol contrast, the calibration decomposition, the mechanistic controls, and the robustness analyses in that order. Section 5 discusses implications and limitations.

2 Related Work

Multi-task ADMET modeling.

Multi-task architectures are common in ADMET prediction. MolP-PC Jiang et al. (2025) trains a multi-view, multi-task framework over 54 endpoints and reports

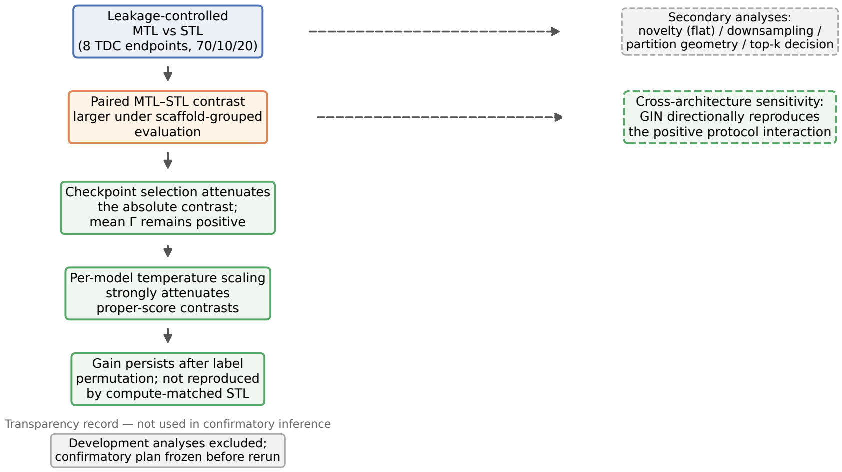

Fig. 1 Study design and principal diagnostic findings. Main evidential chain (top to bottom): leakage-controlled MTL vs. STL comparison reveals a paired contrast that is larger under scaffold-grouped evaluation; checkpoint selection attenuates the absolute contrast while preserving a positive mean protocol interaction; per-model temperature scaling strongly attenuates the proper-score contrasts; label-permuted and compute-matched controls show the gain persists after full label permutation and is not reproduced by additional single-task updates. The matched-scale GIN sensitivity and the secondary analyses (novelty dependence, downsampling, partition-geometry reweighting, decision-level analysis) are independent dashed branches; the development history is a transparency record, not part of the confirmatory evidence. Solid arrows indicate the main evidential chain; dashed arrows indicate parallel secondary and cross-architecture analyses. Development analyses were excluded from confirmatory inference.

that multi-task learning outperforms single-task training on 41 of 54 tasks; MTAN-ADMET Shahid et al. (2025) proposes an adaptive multi-task network over 24 endpoints with task-specific learning dynamics; industrial-scale studies report benefits of multi-task models on time-split data Walter et al. (2024). Reported gains vary across datasets and settings, and task balancing and endpoint size are themselves known to affect transfer in general MTL Kearnes et al. (2016). These studies report average or per-task performance under a single evaluation protocol and do not condition the reported contrast on protocol choice, endpoint scale, or novelty.

Split protocols and out-of-distribution evaluation.

A growing line of work examines how evaluation protocols change conclusions in molecular property prediction. Fooladi et al. (2025) compare 14 model classes across ten split protocols on eight datasets and find that scaffold splits are not uniformly harder than random splits while cluster-based splits are; Guo et al. (2025) reach similar conclusions on cancer cell lines and show that rank correlations between in-distribution and out-of-distribution performance collapse under cluster splits. Zheng et al. (2026) show that structural-frontier splits expose failures hidden by scaffold splits, and Guo and Ding (2026) find that model-size conclusions depend on protocol difficulty. These studies evaluate single-task models; the paired MTL–STL contrast is not their dependent variable, and none manipulates endpoint size or measures continuous target-relative novelty.

Multi-task transfer under realistic splits.

Studies at the intersection of MTL and split protocols are sparse and reach differing conclusions. Kearnes et al. (2016) observed that multi-task gains were modest and dataset-dependent under temporal splits in industrial ADMET data; Wu et al. (2026) found negative transfer in a two-task setting under scaffold splits, mitigated by safe scheduling; Walter et al. (2024) and CheMLT-F Mun and Fazli (2026) report multi-task benefits persisting under realistic splits. These results are not directly comparable because endpoint sets, architectures, metrics, and molecule-holdout schemes differ. A compact comparison of the surveyed studies across eight design axes—architecture-matched STL, evaluation protocols, global cross-task holdout, a paired protocol estimand, checkpoint-rule sensitivity, strict same-model calibration, label-permutation controls, and compute-matched controls—is provided in Additional file 1, Table S1. None of the surveyed studies combines these axes.

Benchmark reliability and evaluation variability.

The molecular machine-learning community has been repeatedly reminded that evaluation choices can dominate model differences. In single-cell genomics, two major benchmarks reached contradictory conclusions about foundation-model perturbation prediction Ahlmann-Eltze et al. (2025); Wei et al. (2026), with the discrepancy traced to evaluation protocol details. Within molecular property prediction, rank correlations between in-distribution and out-of-distribution performance can collapse under cluster splits Guo et al. (2025), which implies that conclusions drawn from a single benchmark realization—including a single split instance and seed—should be treated cautiously. Our study follows this diagnostic tradition: we measure the conditions under which a design choice (task-balanced hard-sharing MTL) delivers or fails to deliver its reported advantage, with multiple split instances and seeds so that evaluation variability is quantified rather than assumed away.

3 Methods

3.1 Estimand hierarchy

For endpoint e, protocol p, split instance k, training seed r, and test set T_{epk}, let $\hat{p}^{\mathrm{STL}}_{iepkr}$ and $\hat{p}^{\mathrm{MTL}}_{iepkr}$ denote the predicted probabilities of the single-task and multi-task models for molecule i. We define the molecule-level log-loss contrast

$$d_{iepkr} \;=\; \ell(y_i, \hat{p}^{\mathrm{STL}}_{iepkr}) - \ell(y_i, \hat{p}^{\mathrm{MTL}}_{iepkr}), \tag{1}$$

where $\ell(y,p) = -y\log p - (1-y)\log(1-p)$; positive d favors the multi-task model. Endpoint-level and protocol-level aggregates are

$$\Delta_{epkr} = \frac{1}{|T_{epk}|}\sum_{i\in T_{epk}} d_{iepkr}, \qquad \Delta_p = \frac{1}{8}\sum_e \mathbb{E}_{k,r}[\Delta_{epkr}], \qquad \Gamma = \Delta_{\mathrm{scaffold}} - \Delta_{\mathrm{random}}. \tag{2}$$

Δ_p is the primary endpoint-macro estimand (endpoints weighted equally), and Γ is the protocol-contrast estimand. We treat the eight endpoints as a fixed target set; we do not claim a superpopulation of all ADMET endpoints. Log loss is a strictly proper measure of probabilistic accuracy; ranking discrimination (AUROC/AUPRC) is assessed separately.

3.2 Models and training

The multi-task model comprises a shared 1024–256–128 encoder (ReLU activations, dropout 0.2 after the first hidden layer) and eight independent affine heads mapping the 128-dimensional representation to one logit per endpoint. Model inputs are fixed-length binary fingerprints computed with RDKit's Morgan algorithm (`GetMorganFingerprintAsBitVect` RDKit contributors (2024)): radius 2, 1024 bits, default settings (`useChirality=False`, `useFeatures=False`), i.e., the standard non-chiral variant commonly denoted ECFP4; fingerprints were computed once per endpoint from the canonical SMILES and reused across all split instances and seeds. Each single-task model is an independently initialized encoder of the same architecture with one corresponding affine head; we therefore compare against *architecture-matched* single-task models, noting that capacity is matched only on the target-task pathway, not globally.

Training uses *equal task sampling with equal loss coefficients*. Each epoch consists of 64 MTL optimizer updates: eight batches of size $B = 512$ drawn with replacement from each of the eight tasks, each update using that task's unweighted binary cross-entropy; uniform scheduling gives equal task weighting in expectation. With 60 epochs, each task's head receives $U = 480$ updates, and the shared trunk receives 3,840 updates in total. Each single-task model receives $U = 480$ updates with $B = 512$ examples under the same sampler; collectively the eight STL models also receive 3,840 updates. Target-task exposure and optimizer-step count are therefore matched between MTL and STL; total examples processed differ (the MTL model additionally consumes auxiliary-task data). Optimization uses Adam ($\eta = 10^{-3}$, weight decay 10^{-4}) with the learning-rate schedule indexed by optimizer step. We scope all conclusions to this task-balanced hard-sharing architecture. As a sensitivity analysis, the protocol contrast was re-estimated with a graph neural network: a three-layer graph isomorphism network (GIN) with hidden width 64, sum pooling (`global_add_pool`), and per-endpoint linear heads, trained on molecular graphs with the same task-balanced schedule (Adam, $\eta = 10^{-3}$, batch size 128, 80 epochs, eight updates per task per epoch) under the same leakage-controlled design and both protocols, at the same scale as the MLP (five split instances $\times$ three seeds). All experiments ran under Python 3.10 with PyTorch 2.13.0 (CUDA 13.0 build) on a single NVIDIA RTX 4070 Ti SUPER GPU; RDKit 2023.09.6 was used for canonicalization, fingerprinting, and scaffold computation.

3.3 Mechanistic controls

Three additional training configurations were evaluated as post hoc mechanistic controls. The label-permuted and pooled-pretraining controls were run under the random protocol at three split instances $\times$ three seeds with all other frozen settings unchanged. (i) *Label-permuted MTL*: for each target endpoint separately, the labels of all other seven endpoints are permuted within each auxiliary endpoint (fixed per-(instance, seed, endpoint) permutations), preserving input exposure, supervision count, and parameterization while removing genuine cross-task label information from the shared trunk; the target-task head still receives true labels. (ii) *Pooled pretrain + STL fine-tune*: a hard-sharing MTL run is used as a shared-trunk initialization, after which each endpoint's head is re-initialized and the full network is fine-tuned on that endpoint alone with the same task-balanced update schedule. These controls compare (i) label transfer against data exposure and (ii) joint-training gains against transferable-representation gains. A further compute-matched control (STL-8$\times$) trains each single-task model on its own true data alone but with the multi-task trunk's full 3,840 optimizer updates (identical batch size, optimizer, and step-indexed schedule), so that optimization budget is matched while auxiliary-task input and labels are absent. The protocol-scale matrix is: label-permuted MTL and pooled pretraining were evaluated under grouped-random allocation at three split instances × three seeds; the compute-matched STL-8× control was evaluated under grouped-random allocation at five instances × three seeds; the label-permuted MTL and STL-8× controls were additionally repeated under scaffold-grouped allocation at three instances × three seeds; pooled pretraining was evaluated only under grouped-random allocation. All controls are post hoc mechanistic diagnostics reported as such; the confirmatory contrasts are unaffected.

3.4 Leakage-controlled evaluation

A test molecule held out from endpoint e may still appear in the training partition of another endpoint, exposing its structure to the multi-task model through auxiliary labels. We control for this *cross-task identity exposure* by global molecule allocation: each molecule is assigned to a single global partition, and every endpoint inherits that partition. Evaluation is therefore inductive with respect to molecule identity across tasks: a test molecule cannot appear in any auxiliary endpoint's training partition. No exact canonical-SMILES identity exposure remains under the primary allocation. Because the pipeline applies no salt removal, tautomer normalization, or charge normalization, we additionally audited identity under a standardized-parent key (RDKit Cleanup, FragmentParent, uncharging, and canonical-tautomer canonicalization; Additional file 1, Section S12). Molecule identity is canonicalized by RDKit canonical-SMILES round-trip (`MolFromSmiles`/`MolToSmiles`) under a fixed RDKit version, which retains defined stereochemistry in the canonical form. Per endpoint, rows that fail canonicalization are dropped, and rows sharing a canonical SMILES are deduplicated keeping the first occurrence; this is the frozen rule for duplicate and conflicting labels, applied identically in the development and confirmatory pipelines. Across the eight endpoints, 13 molecules carry conflicting labels (3 in hERG, 10 in BBB_Martins; none elsewhere); excluding all of them changes the endpoint-macro contrasts by less than 2×10^{-4} under both protocols and both raw and calibrated variants, so the deduplication rule does not drive any reported result. We evaluate two protocols under this scheme: grouped random allocation and Bemis–Murcko scaffold-grouped allocation Bemis and Murcko (1996). Scaffolds are computed by the Murcko algorithm on the canonical SMILES; acyclic molecules, whose scaffold is empty, form a single group (1,630 of 26,638 unique molecules, 6.1%), and molecules that fail SMILES parsing are pooled into a separate reserved group (none occur in this dataset), so every molecule participates in the group-level 70/10/20 allocation. Five split instances and three training seeds are used; split and seed lists, and their hashes, are frozen. Each instance allocates 70% of the unique-molecule pool to training, 10% to calibration, and 20% to test, and this allocation is global: a molecule's partition is identical across all endpoints. The calibration partition exists solely for per-model temperature fitting (Section 3.7) and is disjoint from both training and test within each instance. The confirmatory dataset comprises 33,238 unique molecule–endpoint pairs from 17,972 unique canonical molecules; across the 5×3 (instance, seed) combinations this yields 148,008 test observations, with each (instance, seed) combination containing 9,814–9,930 unique molecule–endpoint pairs and no within-combination duplicates. Cross-instance overlap arises because different split instances draw different test partitions; the identity key is the same canonical SMILES used for global allocation.

3.5 Continuous target-relative novelty

For each test molecule we define *target-relative novelty* as one minus the maximum Tanimoto similarity (ECFP4 Morgan (1965), fixed RDKit version RDKit contributors (2024)) to any molecule in the target endpoint's training partition. Distance to auxiliary-task training molecules is deliberately excluded from the primary definition, so the axis measures distance from the endpoint's own training support; we preregister a secondary quantity for nearest-auxiliary-training similarity and its interaction with target-relative novelty, since a molecule far from target training but close to auxiliary training is precisely a candidate transfer case.

3.6 Endpoint scale and downsampling

Endpoint scale is the number of unique target-task training molecules in each split. With eight endpoints, an observational scale association is weak and confounded by endpoint identity; we therefore prespecify a downsampling experiment: large endpoints are subsampled to a fixed target-training size with a frozen sampler (nested subsets, class prevalence preserved, fixed validation and test partitions, auxiliary-task data unchanged, task-balanced schedule held constant). The defensible conclusion from this experiment is an effect of *target-label availability* within the examined endpoints under this training schedule, not a general "scale rather than identity" claim.

3.7 Calibration and discrimination decomposition

A Brier decomposition explains the Brier-score contrast, not the log-loss contrast. We therefore report (i) a Brier reliability–resolution decomposition as a descriptive complement, (ii) calibration intercept and slope, and (iii) AUROC/AUPRC contrasts. A finding of stable AUROC together with worse Brier reliability is reported as evidence *consistent with* a calibration difference, not as proof that the NLL penalty is primarily calibration.

Temperatures are fitted per run, protocol, model, and endpoint, and strictly within the same trained model. Global molecule allocation now assigns each molecule to one of three partitions: 70% training, 10% calibration, and 20% test (random protocol: molecule-level; scaffold protocol: scaffold-grouped). For each (split instance, training seed) run, one scalar temperature T is fitted by minimizing the mean log loss (NLL) on that same trained model's own calibration partition (disjoint from its training and test partitions; bounded scalar optimization, SciPy `minimize_scalar`, method `bounded`, $T \in [0.2, 5.0]$) and applied to that same model's own test partition; no temperature is shared across models. The calibrated molecule-level contrast is recomputed from temperature-scaled probabilities.

3.8 Checkpoint-selection sensitivity

As a sensitivity analysis at the full five-instance scale, the calibration partition (10%) is split molecule-level into validation (5%) and temperature-calibration (5%) subsets under each protocol (scaffold grouping applies to the allocation of the parent calibration partition). During training, validation NLL is evaluated every ten epochs (the mean over the eight heads for MTL, each head on its own endpoint's validation partition; the endpoint's own validation partition for STL), and model states are retained at every evaluated epoch. Three rules are computed on these checkpoints: fixed-final; global-vs-global (MTL and the eight STL models each select one common epoch minimizing the endpoint-macro validation NLL); and endpoint-vs-endpoint (each MTL head and its corresponding STL select on that endpoint's own validation NLL); a fourth rule matching an earlier post hoc analysis (MTL common epoch, STL endpoint-specific) is reported as an additional sensitivity. Temperature is then fitted on the 5% temperature-calibration subset of the same trained model at the selected checkpoint and applied to its test predictions. The analysis is reported at the split level because the three seeds within a split share a test set.

3.9 Uncertainty and preregistration

The fifteen split-seed combinations are not fifteen independent replicates: three training seeds share a test set, split instances reuse molecules, all endpoints share one MTL model, and a molecule can carry multiple endpoint labels. Confirmatory inference therefore uses paired contrasts with clustering by standardized molecule, distinguishing split variability from training variability; we report split-level and seed-level dispersion separately. The analysis plan—hypotheses, directional tests, split and seed hashes, estimands and weighting, exclusions, downsampling design, novelty model, multiplicity handling, and analysis code version—was frozen in an immutable record (repository commit 5c2cd1e; protocol/final_confirmatory_plan.md) before confirmatory computation on the leakage-controlled design; the development pass (a per-endpoint split subsequently identified as cross-task-exposed) informed the design and is reported separately as a contaminated stress test (Additional file 1, Section S2), excluded from confirmatory statistics.

4 Results

We report results for eight binary ADMET endpoints from the Therapeutics Data Commons Huang et al. (2021): hERG, AMES, BBB penetration, P-glycoprotein inhibition, three cytochrome P450 inhibition tasks, and oral bioavailability (Table 1). All conclusions below are restricted to this endpoint set and to task-balanced hard-sharing MTL.

Under the prospectively frozen fixed-final checkpoint rule, across the leakage-controlled design (global molecule allocation, five split instances, three seeds), the molecule-level log-loss contrast Δ is positive for every endpoint under both protocols: multi-task models assign lower log loss than their architecture-matched single-task counterparts in all 8×2 protocol–endpoint configurations.

Endpoint	Test predictions	Unique molecules	Positive rate
hERG	1,938	443	0.69
AMES	21,933	4,914	0.55
BBB_Martins	5,907	1,322	0.76
Pgp_Broccatelli	3,747	832	0.53
CYP2C9_Veith	36,201	8,132	0.33
CYP2D6_Veith	39,246	8,829	0.19
CYP3A4_Veith	37,161	8,350	0.41
Bioavailability_Ma	1,875	416	0.77

Table 1 Dataset statistics. "Test predictions" counts molecule–endpoint test observations across 5 split instances × 3 seeds (within-(instance, seed) sets contain no duplicates; unique molecules across instances).

4.1 Protocol interaction and overall effect

The endpoint-macro mean contrast is $\Delta_{\text{random}} = +0.321$ under grouped random allocation and $\Delta_{\text{scaffold}} = +0.402$ under Bemis–Murcko scaffold-grouped allocation, with a protocol contrast of $\Gamma = \Delta_{\text{scaffold}} - \Delta_{\text{random}} = +0.081$ (paired $t_7 = 2.65$, $p = 0.033$). Because the eight endpoints and five split instances are fixed in this study, we report the split-instance block bootstrap as the primary uncertainty statement: resampling the five split instances first (then aggregating endpoints within them) gives 95% CI $[+0.048, +0.112]$ ($P(\Gamma > 0) = 1.00$), covering split-instance variability. The molecule-level interval (95% CI $[+0.079, +0.083]$) is conditional on the observed split instances and training runs and answers only "how much sampling noise among molecules," not variability across instance realizations; the endpoint-resample interval (95% CI $[+0.026, +0.138]$; $P(\Gamma > 0) = 0.998$, across-endpoint SD 0.088) covers endpoint heterogeneity; and the paired $t_7 = 2.65$ ($p = 0.033$) and the leave-one-endpoint-out range (+0.069 to +0.096) are reported as sensitivity analyses. The primary and all sensitivity intervals exclude zero, so the positive protocol contrast remained stable under molecule-, split-instance-, and endpoint-level resampling within this benchmark panel. The endpoint-resample interval does not imply an ADMET-endpoint population, because the eight endpoints are a fixed target set. The per-endpoint contrasts Γ_e are positive in six of eight endpoints (Figure 2, panel B); the interaction is concentrated

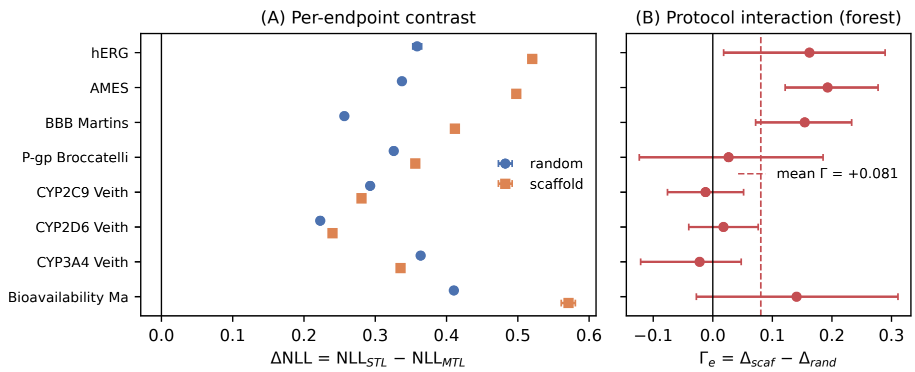

Fig. 2 (A) Per-endpoint log-loss contrast $\Delta = \mathrm{NLL}_{\mathrm{STL}} - \mathrm{NLL}_{\mathrm{MTL}}$ (positive values favor MTL) under grouped random (circles) and scaffold-grouped (squares) allocation, with molecule-level cluster-bootstrap 95% intervals. (B) Per-endpoint protocol contrasts $\Gamma_e = \Delta_{\mathrm{scaf}} - \Delta_{\mathrm{rand}}$ (forest; error bars: split-instance bootstrap 95% intervals, seeds averaged within each split) with endpoint-macro mean (dashed). Six of eight endpoints show positive Γ_e; leave-one-endpoint-out range +0.069 to +0.096.

among the smaller endpoints in this panel (hERG, AMES, BBB, Bioavailability, each $\Gamma_e \approx +0.16$), while the two largest CYP endpoints show $\Gamma_e \approx 0$ or slightly negative (CYP2C9 −0.012, CYP3A4 −0.028). No single endpoint carries the interaction (leave-one-endpoint-out range +0.069 to +0.096), and the paired test is a fixed-set diagnostic rather than a superpopulation claim. The multi-task benefit is therefore *larger under the scaffold protocol* for the smaller endpoints, and the paired MTL–STL contrast is protocol-dependent: moving from grouped-random to scaffold-grouped evaluation need not shrink the observed MTL advantage. We report the contrast on these eight fixed endpoints; we do not claim an ADMET-wide population effect.

4.2 Calibration/discrimination decomposition and robustness

The contrast decomposes into raw-probability and calibration components. Under the random protocol, Brier scores and reliability components improve for all eight endpoints while AUROC is essentially unchanged; under the scaffold protocol, Brier and reliability improve everywhere and AUROC improves in four of eight endpoints (e.g., hERG 0.773 → 0.812). To test whether the log-loss advantage reflects a durable calibration difference or merely better raw probability estimates, we fitted a per-endpoint temperature T Guo et al. (2017) under the strict per-model design: for each (split instance, training seed) run, protocol, model, and endpoint, one scalar temperature was fitted by minimizing the mean log loss on that same trained model's own calibration partition (disjoint from its training and test partitions) and applied to that same model's own test partition; no temperature is shared across models. After this calibration, the mean contrast collapses from +0.321 to −0.003 under the random protocol and from +0.402 to +0.003 under the scaffold protocol, and the calibrated protocol contrast is $\Gamma_{\text{cal}} = +0.006$ (per-endpoint calibrated contrasts positive in five of eight endpoints; paired $t_7 = 0.80$, $p = 0.45$). The uncalibrated gains—including the protocol interaction ($\Gamma_{\text{raw}} = +0.081$ vs. $\Gamma_{\text{cal}} = +0.006$)—are therefore attenuated to near-zero point estimates by per-model temperature rescaling (Figure 3), consistent with differences in raw logit scale rather than with ranking information or a calibration advantage that survives recalibration; we report point estimates and note that a formal equivalence analysis against a practical margin is needed before claiming elimination. The calibration statement is checkpoint-rule specific: under the frozen fixed-final recipe the positive raw contrasts attenuate to near zero, while under per-epoch validation selection calibration substantially reduces but does not eliminate the negative contrasts ($-0.49 \rightarrow -0.28$ random, $-0.45 \rightarrow -0.26$ scaffold); the checkpoint rule sets the direction of the absolute contrast, calibration strongly modulates its magnitude, and the protocol interaction is positive under both rules. A bound-sensitivity check (log-space fitting, $T \in [0.05, 100]$) leaves the calibrated contrasts approximately zero under both protocols (Additional file 1, Section S3). Class-conditional analysis shows the contrast is carried predominantly by positive-class molecules (+0.44 vs. +0.21 for negatives under random allocation).

Checkpoint-selection sensitivity. The fixed final-step recipe is part of the frozen design; we additionally asked whether the headline protocol interaction survives validation-based checkpoint selection, and ran this sensitivity at the full five-instance

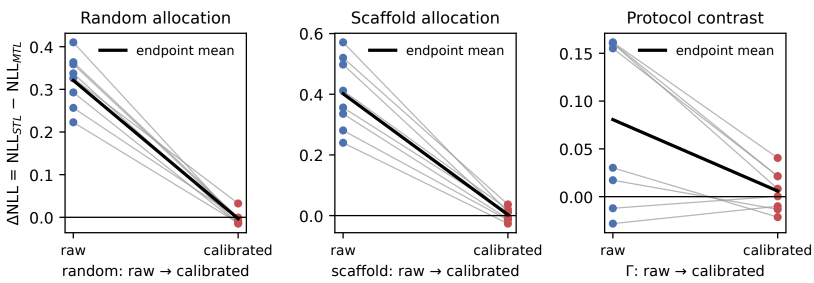

Fig. 3 Temperature-calibration contrast. Mean Δ under both protocols, raw versus strict per-model temperature calibration: the raw advantage (+0.321 random, +0.402 scaffold) attenuates to approximately zero, and the protocol contrast Γ from +0.081 to +0.006 (points: per-endpoint values).

scale under three rules: fixed-final; global-vs-global (MTL and the STL set each select one common epoch minimizing endpoint-macro validation NLL); and endpoint-vs-endpoint (each MTL head and its corresponding STL select the epoch minimizing that endpoint's validation NLL). The rule previously reported post hoc (MTL common epoch, STL endpoint-specific) is retained as an additional sensitivity (Additional file 1, Table S4, Figures S3–S4). The 10% calibration partition is split molecule-level into validation (5%) and temperature-calibration (5%) subsets; under this recipe validation NLL rises monotonically with epoch for both MTL and STL (log loss penalizes the overconfidence that grows with training), so validation-selected rules return early checkpoints. Under both symmetric rules the absolute contrast is attenuated but remains positive ($\Delta_{\mathrm{random}} = +0.185$ and $+0.183$; $\Delta_{\mathrm{scaffold}} = +0.245$ and $+0.255$, versus $+0.321$ and $+0.404$ at the fixed final step), and the protocol interaction remains positive under every rule (split-level $\Gamma = +0.083$, $+0.059$, $+0.072$, and $+0.076$ for fixed-final, global-vs-global, endpoint-vs-endpoint, and the hybrid rule, positive in all five split instances under every rule; split-instance bootstrap 95% intervals in Additional file 1, Table S4). The headline finding—that the paired MTL–STL contrast is larger under scaffold-grouped evaluation—therefore holds across checkpoint rules, while the magnitude of the absolute contrast is modulated by the checkpoint rule: selection attenuates but does not reverse it, and the sign of the contrast is stable under this training recipe. We report the fixed-final rule as primary because it is the prospectively frozen recipe, and the validation-selected analyses as sensitivities (Methods); all analyses share the same train/validation/calibration/test isolation, and the sensitivity is reported at the split level because the three seeds within a split share a test set.

4.3 What drives the contrast: mechanistic controls

The MTL and STL systems differ in representation sharing but also in data exposure and optimizer budget: the MTL shared trunk receives 3,840 optimizer updates (eight tasks $\times$ 480 per-task updates) versus 480 for each STL trunk, and the MTL model processes auxiliary-task molecules the STL model never sees. Three controls under the random protocol isolate these channels (three split instances $\times$ three seeds for the label and pretraining controls; five instances $\times$ three seeds for the budget control; all other frozen settings unchanged). First, a *compute-matched STL* baseline (STL-8$\times$): each endpoint is trained on its own true data alone but with the MTL trunk's 3,840 updates (identical batch size, optimizer, and schedule). STL-8$\times$ is *worse* than standard STL (-0.096 log-loss units; negative in all 15 runs): the multi-task advantage was not reproduced by extending single-task training to the matched 3,840-update budget under the frozen optimizer and learning-rate schedule. (We report the fixed final-step checkpoints; no test-based early stopping was used, so the control answers "same final update budget," not "independently tuned best STL.") Second, a *label-permuted MTL* model: for each target endpoint separately, the labels of all *other* seven endpoints are permuted within each endpoint (fixed per-(instance, seed, endpoint) permutations; identical input exposure, supervision count, and parameterization), so the shared trunk cannot extract genuine cross-task label information while the target head still receives true labels. This fully permuted control leaves the contrast essentially unchanged (+0.296 vs. +0.298 for standard MTL; +0.387 vs. STL-8$\times$), indicating that the raw gain does not require genuine auxiliary-label information and is not reproduced by additional target-task updates. Third, a shared trunk pretrained jointly on all endpoints and then fine-tuned per endpoint (head re-initialized, full-network fine-tuning) retains only a small fraction of the contrast (+0.036, versus +0.298 for hard-sharing joint training), showing that most of the gain requires the joint training objective itself rather than a transferable shared representation alone (Figure 4). The same two decisive controls were repeated under the scaffold protocol (three split instances $\times$ three seeds): label-permuted MTL leaves the scaffold-protocol contrast essentially unchanged (+0.441 vs. +0.407 for standard MTL), and the protocol interaction survives full label permutation: at matched three-instance scale the split-level protocol contrasts are +0.123/+0.069/+0.094 for the real labels (mean +0.095) and +0.114/+0.109/+0.162 for the permuted labels (mean +0.128); the permuted-label interaction was numerically larger in this three-instance post hoc analysis (paired differences −0.009/+0.040/+0.067, mean +0.033, positive in two of three splits), but with only three splits the difference is not statistically distinguishable. The protocol-sensitive gain therefore does not require semantic cross-task label transfer. STL-8$\times$ is again worse than STL under the scaffold protocol (−0.125), confirming that additional single-task updates do not reproduce the gain in either protocol. Together, the controls narrow the plausible explanation to effects specific to concurrent shared-trunk optimization with auxiliary inputs—including auxiliary-input exposure and gradient-noise regularization—rather than semantic cross-task label transfer or additional target-task updates; they do not uniquely identify a single mechanism. Consistent with the main calibration finding, the raw gain of every control configuration is attenuated to near zero by per-model temperature calibration (standard MTL +0.298 $\rightarrow$ −0.001; label-permuted +0.296 $\rightarrow$ −0.016; pooled pretraining +0.036 $\rightarrow$ +0.011; within-run temperature fits on an internal calibration subset): The results are consistent with concurrent shared-trunk training altering logit scale even without meaningful auxiliary labels, and model-specific calibration removes most of the apparent proper-score

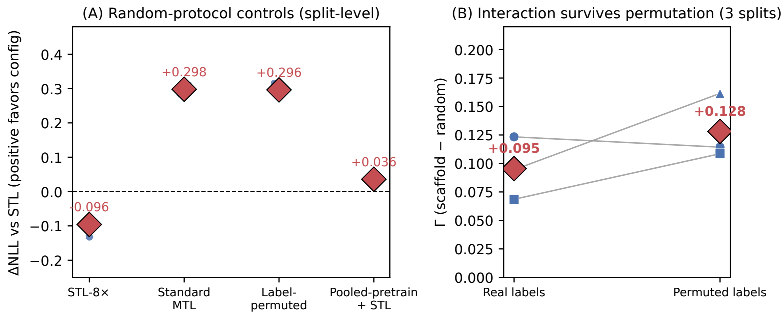

Fig. 4 (A) Mechanistic controls under grouped-random allocation (points: split-level estimates, seeds averaged within each split; three splits for the MTL, permuted, and pooled controls and five for STL-8×; no run-level intervals are shown because seeds within a split share the test set). Compute-matched STL-8× is *worse* than standard STL (−0.096), while standard MTL improves log loss over STL by +0.298, fully permuting all auxiliary labels (target-specific) leaves the contrast essentially unchanged (+0.296), and pooled pretraining followed by per-endpoint fine-tuning retains only +0.036. (B) Protocol interaction under real vs. permuted labels (three-instance matched scale; points: split-level values, seeds averaged within each split; the permuted-label interaction was numerically larger in two of three splits). The protocol-sensitive gain survives full label permutation.

advantage in every configuration. An earlier control that permuted only a fixed subset of auxiliary endpoints produced unstable negative transfer; we attribute that instability to optimization sensitivity of the partially permuted configuration and report only the fully permuted design here. These controls are endpoint-local diagnostics of the main finding.

4.4 Cross-architecture robustness

To test whether the protocol interaction is specific to fingerprint MLPs, we repeated the core comparison with a three-layer graph isomorphism network (GIN) on molecular graphs, under the same leakage-controlled design at matched scale (five split instances, three seeds, architecture-matched STL). The positive direction of the protocol contrast is reproduced: $\Delta_{\text{random}} \approx 0$ (-0.001) and $\Delta_{\text{scaffold}} = +0.025$, giving $\Gamma_{\text{GNN}} = +0.026$ (split-instance block bootstrap 95% CI $[-0.006, +0.064]$; $P(\Gamma > 0) = 0.94$, so at the instance level the GNN interaction is directionally supportive rather than individually significant), although its magnitude and endpoint pattern differ from the fingerprint MLP (positive Γ_e in four of eight endpoints, driven by hERG and Bioavailability).

We read the GNN result as supporting the direction of the interaction in a second representation and model family rather than establishing representation independence (an earlier GNN run with a smaller training budget showed apparent negative transfer; this underfitting artifact and its resolution are recorded in Additional file 1, Section S11).

Endpoint-effect ordering across runs is only weakly consistent (mean pairwise Kendall τ of per-run endpoint-effect rankings: +0.22 under random and +0.33 under scaffold allocation), indicating that conclusions about which endpoints benefit most should not be drawn from a single split realization. Full per-endpoint estimates (contrast, AUROC, AUPRC, Brier, reliability, unique molecule counts) appear in Additional file 1, Table S2.

4.5 Endpoint-scale moderation

Observationally, the contrast is largest for the smallest endpoints under both protocols (e.g., Bioavailability_Ma +0.41; hERG +0.36 under random) and smallest for the largest ones (CYP2D6 +0.22), and the same ordering holds under scaffold allocation (Figure 2). A prespecified downsampling intervention subsampled the three CYP endpoints to 2,000 target-task training molecules under the current 70/10/20 design: the contrast increased in all three (CYP2C9 +0.28 → +0.40; CYP2D6 +0.24 → +0.38; CYP3A4 +0.34 → +0.53), consistent with shared representations helping more when target-task labels are scarcer; because the trunk is shared, non-CYP endpoints also moved (mixed direction), so we do not claim a causal "scale rather than identity" mechanism (full details in Additional file 1, Section S4).

4.6 Continuous novelty analysis

Under the confirmatory design the contrast shows no systematic dependence on target-relative novelty under either protocol: within-endpoint novelty slopes straddle zero for most endpoints, and a hierarchical interaction model yields a detectable but negligible effect ($\hat{\beta} = -0.006$; 95% CI $[-0.010, -0.001]$). The preregistered transfer-window hypothesis is not supported, and an apparent reversal seen in a three-instance development analysis did not survive the five-instance design (Additional file 1, Figures S1 and S2). A partition-geometry control (reweighting random-protocol test molecules to the scaffold novelty distribution) leaves the random-protocol contrast unchanged (+0.3004 vs. +0.3006), so the scaffold advantage is not explained by the between-protocol novelty shift.

4.7 Decision-level implications

A top-k prioritization comparison on the three toxicity endpoints (computed within each run) showed only small and heterogeneous ranking differences between MTL and STL (full results in Additional file 1, Section S7); we do not claim a general prioritization advantage.

5 Discussion

5.1 What the results show

The results qualify, rather than overturn, the commonly reported multi-task advantage in ADMET modeling Jiang et al. (2025); Shahid et al. (2025): under the frozen fixed-final recipe and leakage-controlled evaluation, task-balanced hard-sharing MTL provides better raw probability estimates on all eight endpoints and both protocols. The headline finding is the protocol interaction: the paired MTL–STL proper-score contrast is +0.321 under random allocation and +0.402 under scaffold allocation ($\Gamma = +0.081$; paired $t_7 = 2.65$, $p = 0.033$), and the positive contrast was robust to molecule-, split-instance-, and endpoint-level resampling (split-instance block bootstrap 95% CI $[+0.048, +0.112]$); leave-one-endpoint-out range +0.069 to +0.096, positive in six of eight endpoints and concentrated among the smaller endpoints. Under the confirmatory five-instance design we found no systematic novelty dependence of the benefit under either protocol. The raw-probability advantage—and the protocol interaction itself—attenuates to a near-zero point estimate not distinguishable from zero under the frozen fixed-final rule after strict per-model temperature calibration, in which each temperature is fitted on the same trained model's own calibration partition and applied to its own test partition (calibrated $\Gamma = +0.006$, paired $p = 0.45$), so the gains are consistent with temperature-correctable differences in raw logit scale rather than ranking information or a durable calibration advantage.

5.2 Relation to prior split-protocol and MTL findings

The protocol-dependence finding extends single-task protocol studies Fooladi et al. (2025); Guo et al. (2025); Zheng et al. (2026) to the multi-task contrast, and it helps resolve the apparent tension between studies reporting multi-task benefits under realistic splits Walter et al. (2024); Mun and Fazli (2026) and studies reporting negative transfer Wu et al. (2026); Kearnes et al. (2016): the direction and magnitude of the benefit depend on protocol and on the metric (proper score versus ranking). The temperature-calibration result adds a methodological caution: raw-probability advantages that are strongly attenuated by model-specific recalibration should be reported as such, and calibration diagnostics should accompany proper-score comparisons. The decision-level analysis revealed only small and heterogeneous ranking differences: MTL matched or exceeded STL in most run-level top-k comparisons, but two protocol–endpoint–k combinations favored STL more often, so we do not infer a general compound-prioritization advantage.

5.3 Checkpointing changes the meaning of an MTL advantage

The checkpoint-selection results qualify the fixed-final estimate: validation-based selection returns early checkpoints (validation NLL rises with training under this recipe) and attenuates the absolute contrast—from +0.321/+0.404 to +0.183–+0.185/+0.245–+0.255 under the symmetric rules—but the contrast remains positive, and the mean protocol interaction remains positive under every checkpoint rule examined at the full five-instance scale. Benchmarks should therefore report the checkpoint rule and the validation trajectory alongside any MTL–STL comparison, and should distinguish absolute-contrast claims from protocol-interaction claims; calibration and the checkpoint rule answer different questions about the same numbers.

5.4 Methodological lessons

Two results in this study were design-sensitive in ways that reward explicit reporting. First, a three-instance scaffold analysis showed an apparent reversal of the novelty dependence that disappeared under the five-instance design; the GNN sensitivity analysis similarly showed apparent negative transfer before the training budget matched the MLP configuration. Both episodes illustrate that grouped-allocation studies should verify instance and budget adequacy before interpreting pattern-level claims, and they motivate the frozen protocol and the separation of development from confirmatory analyses adopted here. Second, the strict per-model calibration design—fitting each temperature on the same model's own calibration partition—yields the same qualitative conclusion as an earlier cross-instance calibration shortcut, but the strict design removes a legitimate reviewer objection to the shortcut; we recommend the strict design whenever calibration is part of the evidence.

5.5 Limitations

The endpoint set is limited to eight binary TDC endpoints and one task-balanced hard-sharing fingerprint MLP architecture; other architectures, weighting schemes, and endpoint families may behave differently. The fingerprint featurization ignores stereochemistry (RDKit Morgan fingerprints with `useChirality=False`); chirality-aware featurizations were not evaluated and could matter for endpoints with stereoselective effects such as hERG and the CYP family. The downsampling manipulation increased the contrast consistently across the three CYP endpoints, but because the multi-task model shares its trunk, the intervention also altered the auxiliary endpoints; the result therefore supports target-label availability as a moderator within this training configuration but does not isolate a general causal sample-size mechanism. Endpoint-effect ordering is unstable across split instances, and with eight endpoints the protocol interaction, while supported by the paired test and bootstrap intervals at all three resampling levels, rests on a small number of endpoint-level observations; we do not claim an ADMET-wide population effect. The temperature-calibration analysis is a point-estimate attenuation analysis; a formal equivalence analysis against a practical margin would be needed to claim elimination, and the calibrated point estimates should be read as approximately zero rather than as proof of exact zero. The top-k decision analysis is endpoint-local and descriptive. We treat all findings as benchmark-local.

6 Conclusion

We presented a leakage-controlled diagnostic with a frozen confirmatory plan of task-balanced hard-sharing multi-task ADMET models. Under the frozen fixed-final checkpoint rule, the multi-task proper-score contrast is positive on all eight endpoints under both protocols and larger under scaffold-grouped evaluation; a validation-selected checkpoint sensitivity attenuates the absolute contrast under both protocols while the protocol interaction remains positive under every checkpoint rule at the five-instance scale. Under the fixed-final rule the contrast is $\Gamma = +0.081$ with split-instance 95% CI $[+0.048, +0.112]$ (positive in six of eight endpoints, robust to molecule- and endpoint-level resampling), corresponding to raw-probability differences that attenuate to a near-zero point estimate after strict per-model temperature calibration under the frozen fixed-final rule (calibrated $\Gamma = +0.006$, $p = 0.45$). Under the confirmatory five-instance design we found no systematic novelty dependence of the benefit under either protocol; an apparent reversal in a three-instance development analysis did not survive the larger design.

Compute-matched and label-permutation controls (under both protocols) indicate that the raw gain is not reproduced by additional single-task updates and does not require genuine auxiliary-label structure—the protocol interaction survives full label permutation—and the positive direction of the protocol interaction was reproduced in a matched-scale GNN sensitivity analysis ($\Gamma_{\text{GNN}} = +0.026$), though at the split-instance level the GNN interval includes zero (95% CI $[-0.006, +0.064]$) and we read the result as directionally supportive rather than individually significant. These results suggest that multi-task proper-score advantages in ADMET modeling should be reported conditionally on evaluation protocol and accompanied by calibration diagnostics, and that instance count in grouped-allocation studies should be large enough to stabilize group sampling. Natural next steps are to extend the diagnostic to larger endpoint panels, alternative multi-task formulations, and additional OOD protocols such as cluster or temporal splits.

Declarations

Availability of data and materials

The dataset(s) supporting the conclusions of this article are available in the Therapeutics Data Commons repository (https://tdcommons.ai/), endpoint identifiers: hERG, AMES, BBB_Martins, Pgp_Broccatelli, CYP2C9_Veith, CYP2D6_Veith, CYP3A4_Veith, Bioavailability_Ma. The analysis code, global molecule-partition split manifests, prediction tables, temperature-calibration module, and the full analysis protocol are available in the GitHub repository https://github.com/Functionhx/

ADMET-MTL-Diagnostic (MIT license; frozen submission snapshot: release v1.0.3). Archived version with persistent DOI: https://doi.org/10.5281/zenodo.21783701.

Competing interests

The authors declare that they have no competing interests.

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Authors' contributions

Y.F. conceived the study, designed the experiments, performed the analysis, and wrote the manuscript.

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Use of AI tools

AI-assisted tools were used for writing support and code assistance during the preparation of this manuscript; all experiments, analyses, and conclusions were performed and verified by the author.

List of abbreviations

- ADMET: Absorption, distribution, metabolism, excretion, and toxicity
- AUPRC: Area under the precision-recall curve
- AUROC: Area under the receiver operating characteristic curve
- ECFP4: Extended-connectivity fingerprint with diameter 4 (Morgan radius 2)

- GIN: Graph isomorphism network
- MTL: Multi-task learning
- NLL: Negative log likelihood
- OOD: Out of distribution
- STL: Single-task learning
- TDC: Therapeutics Data Commons

Additional file

Additional file 1. File format: PDF. Title: Supporting Information for "Evaluation Protocol, Checkpointing, and Calibration Shape Apparent Multi-Task Gains in ADMET Prediction". Description: comparison with prior studies, complete endpoint-level results, calibration diagnostics, checkpoint-selection sensitivity, downsampling analyses, mechanistic controls, decision-level analyses, cross-architecture sensitivity, chemical-space diagnostics, and development history.

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